Data Mining COSC 3125

Assignment 1: Practical Data Mining

Part 1: Classification

Run No	Classifier	Parameters	Training Error (%)	Cross-validation Error (%)	Over-fitting (CV- T)%
1.	ZeroR	default	37.5	37.5	0.0
2.	OneR	default	7.5	8	0.5
3.	J48	default (C=0.25, M=2)	0.5	1.0	0.5
4.	**IBK**	default (k=1)	2.5	4.25	1.75

1.ZeroR gave the weakest performance (37.5% error) as it only predicts the majority class, serving as a baseline. OneR improved accuracy substantially (≈92%) with minimal overfitting, showing that a single attribute is highly predictive. J48 achieved the best results (≈99% accuracy, 1% CV error) with negligible overfitting, effectively capturing the decision structure. IBk (k=1) also performed strongly (≈96% CV accuracy) but showed slightly higher overfitting due to memorization of training instances. Overall, J48 generalizes best, OneR highlights strong single-attribute influence, IBk balances accuracy with some overfitting risk, and ZeroR confirms the baseline.

Run No.	Classifier	Parameters (C, M)	Training Error (%)	CV Error (%)	Overfitting (%)
3	J48	C=0.25, M=2 (default)	0.5	1.0	0.5
5	J48	C=0.10, M=2	0.5	1.0	0.5
6	J48	C=0.05, M=5	0.5	1.0	0.5
7	J48	C=0.50, M=2	0.5	1.25	0.75
9	J48	C=0.05, M=10	0.5	0.75	**0.25**

2. Tuning J48 with different confidence (C) and minimum leaf size (M) values confirmed that the classifier is already highly robust on this dataset. The default setting (C=0.25, M=2) and stronger pruning configurations (C=0.1, M=2 and C=0.05, M=5) all achieved ~99% accuracy with only 0.5% overfitting, while more relaxed pruning (C=0.5) or larger leaves (M=10) slightly increased error. The lowest gap (0.25% overfitting) was observed at C=0.05, M=10, but overall, no parameter combination substantially outperformed the default. This confirms that J48 generalizes extremely well, with consistently high accuracy and negligible overfitting.

3. As the training percentage decreases, test error rises because the model has fewer examples to learn patterns. With 90–66% training, J48 achieved 100% accuracy, but at 50% and 30% splits accuracy fell to 97.5% and 95.4%, confirming that smaller training sets weaken generalisation. This demonstrates that reducing training size increases test error and weakens stability, as the tree becomes less representative of the full dataset.

Run No.	Split (Train/Test)	Train Size	Test Size	Accuracy (%)	Error (%)
1	90 / 10	360	40	100.0	0.0
2	80 / 20	320	80	100.0	0.0
3	66 / 34	264	136	100.0	0.0
4	50 / 50	200	200	97.5	2.5
5	30 / 70	120	280	95.4	4.6

4.

Run No.	Classifier	Parameters	Training Error (%)	CV Error (%)	Overfitting (%)
1	IBk	k=1	4.25	5.25	1.0
2	IBk	k=2	2.25	3.75	1.5
3	IBk	k=3	4.25	5.0	**0.75**
4	IBk	k=5	6.0	7.5	1.5
5	IBk	k=9	11.5	13.5	2.0

IBk with low k (1–2) gave high accuracy but moderate overfitting, while high k (9) underfit with larger errors. The best balance occurred at k=3, where training and CV errors were almost equal and overfitting was lowest (0.75%). Thus, moderate k values (3–5) are most efficient, offering strong accuracy with minimal overfitting.

5.

Run No	Classifier	Parameters	Training Error (%)	CV Error (%)	Overfitting (%)
19	NaiveBayes	default	5.25	5.0	-0.25
20	NaiveBayes	useKernelEstimator=True	2.25	2.0	-0.25
21	RandomForest	numTrees=50	0.0	0.0	0.0
22	RandomForest	numTrees=100	0.0	0.0	0.0
23	RandomForest	numTrees=200	0.0	0.0	0.0

Among the additional classifiers tested, RandomForest achieved perfect performance (0% training and CV error) across all runs, making it the best classifier in terms of predictive accuracy and generalisation. Naïve Bayes also performed strongly (95–98% accuracy), with the kernel estimator variant improving results slightly, and even showing negative overfitting (CV error lower than training error), which indicates good stability.

6.

Classifier	Correctly Classified (%)	Incorrectly Classified (%)	Accuracy
ZeroR	62.5% (250/400)	37.5%	**62.5%**
OneR	92.5% (370/400)	7.5%	**92.5%**
J48	99.5% (398/400)	0.5%	**99.5%**

ZeroR performed worst as a baseline (62.5% accuracy), OneR improved substantially (~92%), and J48 achieved the highest accuracy (~99%) with minimal overfitting. This shows that while a single attribute can predict well, J48's ability to model multiple attribute interactions makes it the most reliable classifier.

7. The key finding was that hemoglobin is a strong predictor, with OneR reaching ~92% accuracy, underscoring its clinical relevance. Moreover, J48 and RandomForest achieved near-perfect accuracy with negligible overfitting, confirming the dataset's strong separability. Finally, IBk with moderate k (3–5) gave the best balance of accuracy and generalisation, showing that careful parameter choice improves stability.

8.

Selected attributes (17 out of 25):

- bp, sg, al, rbc, bgr, bu, sc, sod, pot, hemo, pcv, wbcc, htn, dm, appet, pe, ane

Attribute Set	**No. of Attributes**	**Training Accuracy (%)**	**CV Accuracy (%)**
Full set	25	99.5	99.0
Reduced set	17 (CfsSubsetEval + BestFirst)	91.2	81.2

Applying attribute selection reduced the feature set from 25 to 17 attributes, but accuracy declined sharply from ~99% with the full set to ~81% under cross-validation. This shows that the excluded attributes contributed important predictive information. In this dataset, attribute selection did not improve performance; instead, retaining all features yielded the most reliable and accurate classification.

Part 2: Numeric Prediction

Run No	Model	MAE (Train)	RMSE (Train)	MAE (10-fold CV)	RMSE (10-fold CV)
1	ZeroR	87.38	154.39	87.66	154.76
2	IBk (k=1)	**0.00**	**0.00**	20.83	53.64
3	M5P	12.93	25.65	**13.69**	**35.30**

1. ZeroR serves only as a weak baseline with uniformly large errors. IBk (e.g., k=1) memorizes the training set (MAE=0) but generalizes poorly, with high CV error and large misses on extreme CPUs, showing clear overfitting. In contrast, M5P achieves the best generalization, with the lowest cross-validated MAE/RMSE and more balanced residuals. Its linear models, split mainly by MMAX, capture the dataset's structure better than nearest-neighbour copying. Output predictions confirm this: ZeroR errors are uniformly large, IBk produces big outliers, while M5P maintains smaller, more even errors.

2.

Run No	Classifier	Parameters	Training MAE	CV MAE	Overfitting (CV – Train)
4	M5P	M=2 (small leaves)	12.93	15.97	3.04
5	M5P	M=10 (larger leaves)	14.76	13.69	-1.07
6	M5P	Unpruned, M=4	7.51	11.47	3.96
7	IBk	k=3	18.13	33.03	14.90
8	IBk	k=5, distance-weighted	25.71	36.81	11.10

The results confirm that M5P outperforms IBk for numeric prediction. IBk models showed high errors and severe overfitting, while M5P achieved lower errors with performance shaped by leaf size and pruning. Unpruned and small-leaf models overfit, but M5P with M=10 gave the best balance, producing the lowest CV error (13.7) with negligible overfitting, making it the most accurate and generalizable configuration.

3.

Run No	Classifier	Parameters	Training MAE	CV MAE	Overfitting
9	LinearRegression	Attribute selection = M5 method	28.40	36.78	8.38
10	LinearRegression	Attribute selection = Greedy method	29.02	37.76	8.74
11	SMOreg	Poly kernel, C=1.0	17.45	20.49	3.04
12	SMOreg	Poly kernel, C=2.0	17.47	20.53	3.06
13	SMOreg	RBF kernel, C=2.0, γ=0.01	19.34	20.93	1.59

Linear Regression gave the poorest results, with high CV errors (~37) and large overfitting, showing it cannot model the dataset's complexity. SMOreg performed much better, with CV errors around 20–21. Polynomial kernels were accurate but slightly overfit, while the RBF kernel (C=2, γ=0.01) achieved the best trade-off, with low error (20.93) and minimal overfitting (1.59). Thus, SMOreg with RBF kernel is the most reliable model, combining accuracy with strong generalization.

4. A key finding was the contrast between IBk and M5P: IBk achieved zero training error but severely overfit, while M5P, especially with pruning (M=10), achieved the lowest CV error and best generalization. Another insight was that SMOreg clearly outperformed Linear Regression, showing the importance of non-linear kernels for this dataset. Among them, the RBF kernel provided the best balance of accuracy and generalization, making it the strongest alternative to M5P.

Overall, the golden nuggets are that M5P (pruned, M=10) is the most accurate and interpretable model, while SMOreg with RBF kernel offers a robust non-linear option, both far surpassing simpler or overfitting approaches.

Part 3: Clustering

1. With K=1, all instances formed a single uninformative cluster. At K=2, the data split into lower-credit younger clients and higher-credit older ones. K=3 revealed clear low, medium, and high exposure groups, while K=4 isolated a very high credit/long-duration cluster. Beyond this (5, 10, 20), clusters fragmented into small groups, lowering interpretability despite reduced error. Overall, 3–4 clusters best balance accuracy and meaningful patterns in credit amount, duration, and age.

2. When K-means was run with k=3 but different seeds (20, 42, 70, 100), the initial centroids varied, leading to changes in cluster sizes (e.g., 25%-21%-55% vs. 20%-63%-18%) and WCSS (~237–411). This shows K-means is sensitive to initialization, often converging to different local optima, so results are broadly similar in structure but differ in detail.

3. Running EM with default parameters produced 7 clusters (4%–24%), mainly separated by credit amount, duration, and age, with job type refining profiles. For example, one cluster (24%) had short, small loans with mostly unskilled/skilled workers, while another (11%) captured large, long-term loans with higher-qualified clients. With a log likelihood of –17.07, EM provided a reasonable fit and revealed distinct borrower risk segments.

4. Before normalisation, EM produced 7 clusters dominated by credit_amount due to its larger scale, giving a weaker fit (log likelihood –17.07). After normalisation, EM again found 7 clusters but with a much better fit (0.9855), as duration, credit_amount, and age contributed more evenly. Normalisation prevents scale bias, ensuring fairer attribute weighting and more interpretable, reliable clusters.

5. Changing these parameters showed that minStdDev had the greatest impact: higher values gave fewer, broader clusters, while lower values produced more detailed but riskier (overfitted) ones. In contrast, minLogLikelihoodImprovementCV and Iterating mainly affected convergence speed and stability, not cluster structure. Thus, tuning minStdDev is key for meaningful results, while log-likelihood settings control efficiency.

6.The EM results suggest around seven clusters. One cluster groups older borrowers with lower loan amounts and shorter durations, mostly in stable skilled jobs. This paints them as financially conservative, lower-risk clients, clearly distinct from younger groups seeking higher credit. This shows the algorithm is capturing meaningful socio-economic profiles.

7. K-means with k=2 produced only two broad groups mainly by loan size and age, giving coarse partitions. EM automatically found seven clusters with distinct profiles across credit amount, duration, age, and job, offering richer and more interpretable borrower segments. Overall, EM is better suited for this credit data, while K-means is simpler but less informative.

8. Clustering revealed distinct borrower groups: K-means split low-loan younger clients from older high-loan ones, while EM found seven nuanced profiles. Normalisation improved balance across attributes, making these clusters valuable insights for credit risk and lending decisions.

Task 4: Association Finding

1. bakery-data1.arff represents each transaction in a single row with multiple item attributes (wide format), while bakery-data2.arff represents transactions in long format, where each row contains a transaction ID and a single purchased item. Thus, data1 is compact but harder for association rule mining, whereas data2 is more verbose but directly suited for algorithms like Apriori.

2. When Apriori was run on *bakery-data1.arff* with default settings (support = 0.1, confidence = 0.9), the rules generated were trivial "absence ⇒ absence" associations, such as *Apple Pie=no ⇒ Almond Bear Claw=no*. These appeared statistically strong (confidence ≈ 0.98) but were uninformative because they only reflected the rarity of purchases rather than meaningful co-buying behavior, as confirmed by lift ≈ 1. Lowering support to 0.05 and confidence to 0.8 allowed more purchase-related rules to surface, showing that very high thresholds produce only strict but trivial patterns, whereas moderate values broaden coverage and reveal stronger, interpretable associations like product co-purchases.

3. On *bakery-data2.arff*, Apriori produced no rules under default settings (support = 0.1, confidence = 0.9). This happened because the dataset uses a sparse format, where only purchased items are recorded ("yes") and all absences are left blank, so few items meet the high support threshold. By lowering support to 0.05 and confidence to 0.8, some frequent itemsets were detected, but still only a limited number of rules emerged. This highlights that for sparse data, default thresholds are too restrictive, and meaningful associations can only be uncovered by relaxing support and confidence, though the trade-off is fewer and weaker patterns overall.

4. Compared with Apriori, both alternative associators struggled under default parameters: FilteredAssociator did not reduce trivial rules for bakery-data1, and FPGrowth failed to generate rules for bakery-data2. This highlights that the choice of algorithm and parameter tuning are highly dependent on the data representation — dense binary vs. sparse transactions and meaningful associations only emerge when thresholds are adjusted appropriately.

5. After tuning parameters, a few meaningful co-purchase patterns were discovered, such as *Apple Pie ⇒ Cherry Tart* and *Chocolate Cake ⇒ Coffee Éclair*. While not surprising, these associations are valuable because they highlight common buying habits that could support cross-selling or bundling strategies. In contrast, most default runs produced trivial absence rules or no rules. Thus, the "golden nuggets" lie in the tuned results, where interpretable and actionable rules emerged.

6. With default settings, Apriori produced no meaningful associations in the chronic-kidney-disease dataset (400 instances, 25 attributes). As the output showed, *"No large itemsets and rules found"*, the dataset is too small and thresholds too strict; useful rules would only appear if support and confidence were lowered.

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